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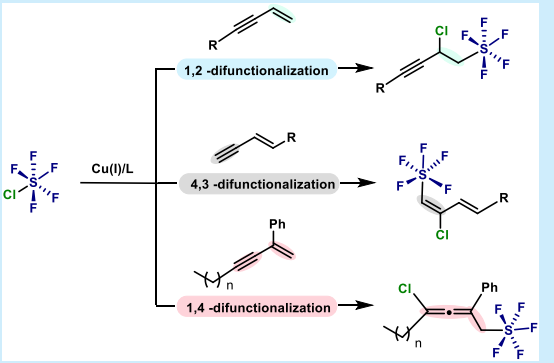
Copper-Initiated Regiodivergent Chloropentafluorosulfanylation of 1,3-Enynes under Substrate Control

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ABSTRACT: A copper-catalyzed regiodivergent chloropentafluorosulfanylation strategy for 1,3-enynes using SF_5Cl has been developed. The regioselectivity is dictated by the structural and substitution patterns of 1,3-enynes, enabling facile access to three classes of SF_5-containing products: propargylic chlorides, 1,3-dienes, and allenes. The reaction systems involve radical species, where the transfer of a chlorine atom from SF_5Cl to a carbon radical is considered the predominant pathway. Diverse types of SF_5- building blocks can be synthesized through simple functional group transformations.



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The pentafluorosulfanyl group (SF_5), a bioisostere of trifluoromethyl group, possesses high electronegativity, exceptional lipophilicity, superior chemical stability, and strong metabolic robustness[1] owing to its distinctive square pyramidal geometry and abundant fluorine atoms.[2] Besides enhancing drug potency,[3] the incorporation of the SF_5 group also confers benefits on the optoelectronic properties of materials[4] and the activity of the parent catalysts[5] (Figure 1A). Therefore, the growing demand for diversified multifunctional SF_5-containing structures necessitates the development of practical and efficient synthetic methodologies. The state-of-the-art synthetic strategy for access to aryl-SF_5 motifs involves the stepwise oxidative fluorination of sulfur(II) precursors.[6] Alternatively, direct pentafluorosulfanylation reactions allowed synthesis of aliphatic pentafluorosulfanylated compounds.[7]

Pentafluorosulfanyl chloride (SF_5Cl) is widely employed as a reagent for direct pentafluorosulfanylation due to its optimal balance between reactivity and stability.[8] Following the seminal report of a user-friendly, gas-reagent free synthetic protocol by Togni,[9] with subsequent contribution by Qing,[10] this reliable source of SF_5 radical has become more manageable and practical. The predominant method for introducing a SF_5 group involves the reactions of SF_5Cl with alkenes or alkynes via atom-transfer radical addition (ATRA) in the presence of initiators such as BEt_3.[11] In addition, it has been discovered that SF_5Cl exhibits reactivity toward α-diazo carbonyl compounds, leading to the formation of α-SF_5 ketones and esters[10,12] (Figure 1B). However, strategies for direct C−SF_5 bond formation remain limited to substrates containing only one reactive site. In contrast, reports on reactions involving more complex substrates featuring multiple reactive sites are scarce,[13] where both regio- and chemoselectivity emerge as key issues.

1,3-Enyne, a common molecule featuring conjugated double and triple carbon−carbon bonds, exhibits multiple reactive sites and can undergo diverse regioselective coupling reactions such as 1,2-, 3,4-, and 1,4-functionalization.[14] Notably, the radical addition strategy for difunctionalization of 1,3-enynes via single-electron transfer offers significant advantages including mild reaction conditions and excellent tolerance toward functional groups.[15] Therefore, we designed a unified protocol for accessing various valuable SF_5-functionalized products through the radical addition of SF_5Cl to 1,3-enynes. While conceptually simple, the implementation of this protocol remains challenging. First, the steric hindrance and unique electronic properties of SF_5 radical should be compatible with the reactivity of 1,3-enynes; second, effective control over the regioselectivity arising from multiple reactive sites in the alkenyl and alkynyl positions remains unclear. Herein we report on copper-initiated regiodivergent pentafluorosulfanylation of 1,3-enynes, which enables access to a wide range of SF_5-substituted propargylic derivatives, 1,3-dienes, and allenes, where the regioselectivity is dictated by the intrinsic properties of the enyne substrates (Figure 1C). Moreover, the resulting SF_5-containing building blocks can be further functionalized

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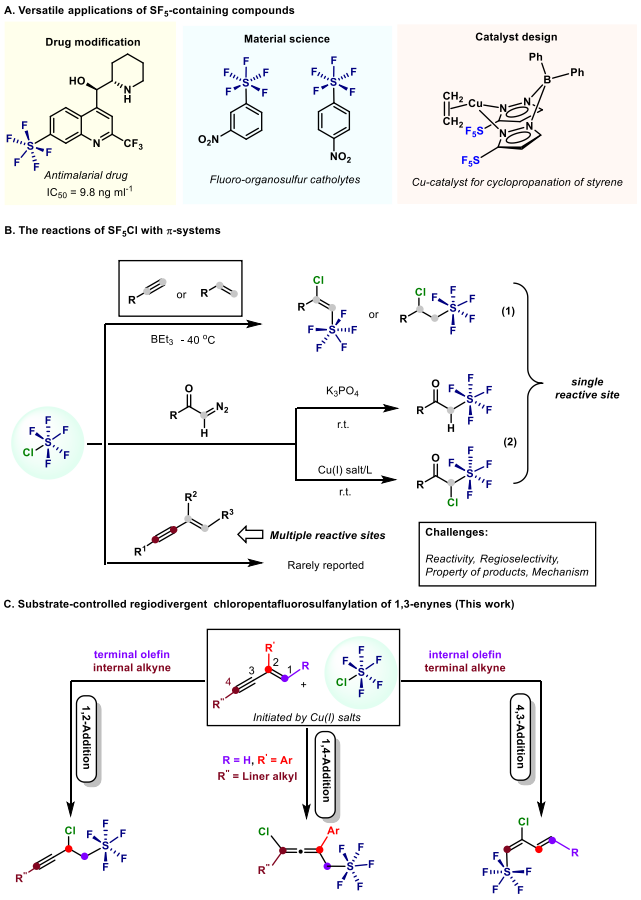


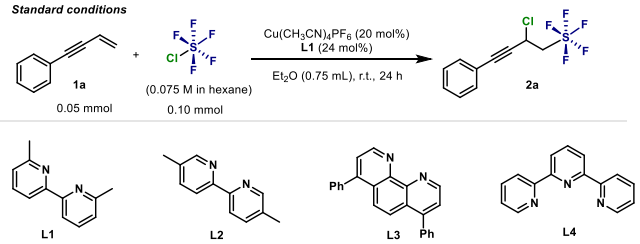
Figure 1. Application and synthesis of SF_5-containing compounds.

through hydrodechlorination, hydrostannation, and dehydrohalogenation.

We commenced our studies of chloropentafluorosulfanylation using but-3-en-1-yn-1-ylbenzene (**1a**) as a model substrate. Considering previous reports on metal-catalyzed difunctionalization of 1,3-enynes,[14] we employed copper catalysis to generate a SF_5-radical and to initiate the reaction. Following extensive optimization studies (see Supporting Information), we eventually identified the optimal conditions (Table 1, entry 1), under which **1a** (1.0 equiv) reacted with SF_5Cl (0.075 M in hexane, 2.0 equiv), in the presence of $Cu(CH_3CN)_4PF_6$ (20 mol %)/6,6′-dimethyl-2,2′-bipyridine (**L1**, 24 mol %) as a catalyst in diethyl ether (0.75 mL) at room temperature for 24 h, resulting in a high yield of the desired product **2a** (90%) with excellent 1,2-selectivity as determined by in situ ^{19}F NMR analysis.

During the optimization process, several noteworthy observations were made. (1) The reactivity of the cationic Cu(I) salt coordinated with a bipyridyl ligand outperformed that of other Cu salts with a terpyridyl- or phenanthroline-type ligand (Table 1, entries 2−6). (2) The addition of 0.75 mL of diethyl ether proved to be indispensable, as deviations from its loading caused detrimental effects on the reaction outcome (Table 1, entries 7 and 8). Moreover, lower yields of **2a** were obtained when other solvents such as THF or dichloroethane were used (Table 1, entries 9−11). This observation implies that the specific Et_2O to hexane ratio facilitates the compatibility of SF_5 radical with the reactivity of 1,3-enyne. (3) By employing the well-established Et_3B-initiated conditions,[11e] only a poor yield was achieved for the 1,2-addition product (Table 1, entry 12).

Table 1. Optimization of the Reaction Conditions for Chloropentafluorosulfanylation of 1,3-Enyne (1a)[a]



Entry	Variations from standard conditions	Yield (%)[b]
1	none	90
2	CuCl (20 mol %) instead of $Cu(CH_3CN)_4PF_6$	12
3	$Cu(OAc)_2$ (20 mol %) instead of $Cu(CH_3CN)_4PF_6$	<5
4	**L2** (24 mol %) instead of **L1**	78
5	**L3** (24 mol %) instead of **L1**	43
6	**L4** (24 mol %) instead of **L1**	0
7	Et_2O (0.25 mL) as cosolvent: Hexane/Et_2O (5.3:1)	47
8	Et_2O (1.0 mL) as cosolvent: Hexane/Et_2O (1.3:1)	53
9	DCE (0.75 mL) instead of Et_2O as cosolvent	21
10	THF (0.75 mL) instead of Et_2O as cosolvent	0
11	Hexane (0.75 mL) instead of Et_2O as solvent	36
12[c]	BEt_3 (10 mol %)/air instead of Cu salt/L	19

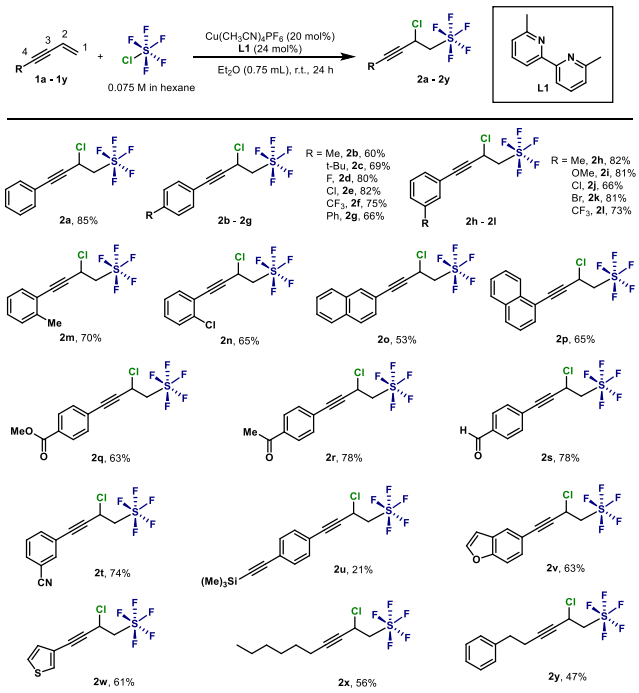
[a]Standard conditions: **1a** (0.05 mmol), SF_5Cl (0.075 M in hexane, 0.1 mmol), $Cu(CH_3CN)_4PF_6$ (0.02 mol), **L1** (0.024 mol) in 0.75 mL of Et_2O [Hexane/Et_2O (1.8:1)] at rt for 24 h under N_2. [b]Yields were determined by in situ ^{19}F NMR analysis using Ph−CF_3 as an internal standard. [c]Reaction conditions: **1a** (0.05 mmol), SF_5Cl (0.075 in hexane, 0.1 mmol), were sealed under N_2. BEt_3 (1 M in hexane, 10 mol %) and 1 mL of air was then injected at −40 °C. Reaction at −40 °C for 24 h.

With the optimized conditions in hand, we proceeded to investigate the scope of this chloropentafluorosulfanylation system using a series of 4-substituted 1,3-enynes (Scheme 1). These substrates, possessing a terminal olefin moiety, exhibited exclusive 1,2-selectivity and delivered the corresponding SF_5-substituted propargylic chlorides in high yields. Notably, both the substitution pattern and the electronic nature of the aryl groups exerted negligible influence on either the yields or regioselectivity (**2a**−**2p**). This protocol also demonstrated excellent compatibility with typical functional groups such as ester (**2q**), ketone (**2r**), aldehyde (**2s**), and cyano (**2t**) groups. Among the examples examined, only a substrate bearing a *para* alkynyl group reacted with low conversion (**2u**), albeit with consistently excellent regioselectivity. Heteroaryl substituents such as a thienyl or benzofuryl were also well tolerated (**2v** and **2w**). Moreover, the alkynyl terminus was also smoothly extended to different alkyl groups (**2x** and **2y**).

To further investigate the scope of the enyne substrates, we employed 4-unsubstituted enynes, which exhibited distinct regioselectivity (Scheme 2). Under identical reaction conditions as in Scheme 1, (*E*)-but-1-en-3-yn-1-ylbenzene (**3a**) selectively coupled with 4,3-regioselectivity to afford the **4a** in 73% yield. Additionally, minor side-products resulting from dichlorination at positions 3 and 4 were also observed. The observed outcomes highlight the significant steric effect around the reactive sites influencing regioselectivity. Subsequent investigation using various 4-unsubstituted 1,3-enynes revealed that both electron-donating and electron-withdrawing groups

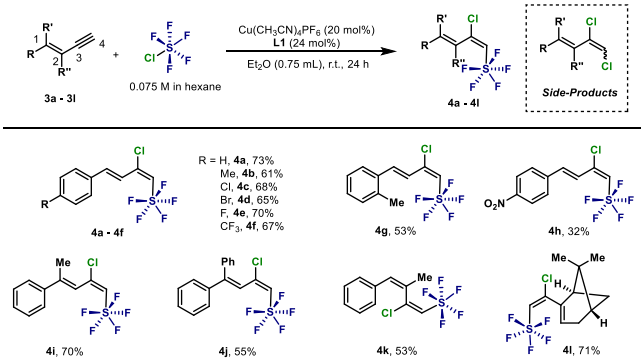
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Scheme 1. Reaction Scope of 4-Substituted 1,3-Enynes[a,b]



[a]Reaction conditions: **1** (0.05 mmol), SF_5Cl (0.075 M in hexane, 0.1 mmol), $Cu(CH_3CN)_4PF_6$ (0.02 mol), **L1** (0.024 mol) in 0.75 mL of Et_2O at rt for 24 h under N_2. [b]Isolated yields.

Scheme 2. Reaction Scope of 4-Unsubstituted 1,3-Enynes[a,b]

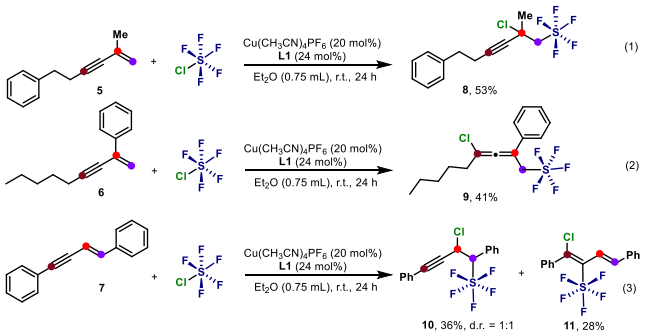


[a]Reaction conditions: **3** (0.05 mmol), SF_5Cl (0.075 M in hexane, 0.1 mmol), $Cu(CH_3CN)_4PF_6$ (0.02 mol), **L1** (0.024 mol) in 0.75 mL of Et_2O at rt for 24 h under N_2. [b]Isolated yields.

on 1-aryl-substituted enynes led to good yields of 1,3-diene formation (Scheme 2, **4b**−**4g**). However, introduction of a *para* nitro group into the aryl ring resulted in a lower yield for this reaction (**4h**, 32%), indicating sensitivity of these substrates toward electronic perturbation. The steric effect also governs the mutual *trans* orientation between the SF_5 and the Cl group, which was confirmed by $^1H-^1H$ NOESY. Intriguingly, both 1,1- and 1,2-disubsituted enynes (**3i**−**3k**), including a cyclene-containing enyne (**3l**), underwent smooth transformation into the desired 4,3-addition products (**4i**−**4l**).

Having observed the pronounced substrate-dependent selectivity, we proceeded to investigate the chloropentafluorosulfanylation reaction of a series of 2,4- and 1,4-disubstituted 1,3-enynes with different substituents (Scheme 3). While a methyl group at the 2-position had minimal impact on the

Scheme 3. Reaction Scope of Multisubstituted 1,3-Enynes



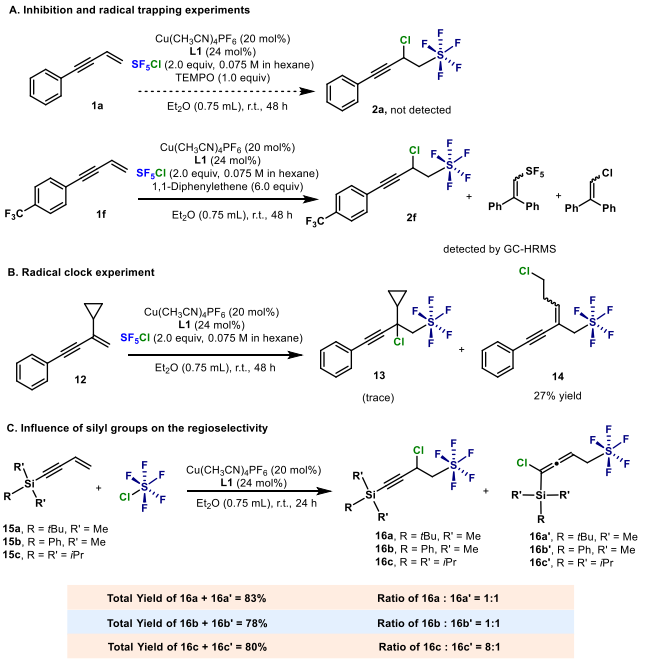
formation of 1,2-difunctionalization product **8**, bulkier groups such as phenyl caused a remarkable switch toward 1,4-selectivity (Scheme 3, reaction 2). Consequently, enyne **6** containing a phenyl and a linear alkyl at the 2- and 4-position respectively reacted through the 1,4-addition pathway, resulting in new-style SF_5-substituted allene **9** as the exclusive F-containing product. In this case, formation of a propargyl radical with subsequent radical isomerization[16] may occur, and the relatively bulky steric substituent disfavors the radical chlorine atom transfer to the 2-position carbon. Regarding 1,4-disubstituted enyne substrate **7**, both propargyl chloride (**10**) and diene product (**11**) were obtained in a 1.3:1 ratio, which suggests that the substituents at the 1-position decreased the regioselectivity of 1,2- versus 3,4-addition.

Experimental studies have been conducted to gain mechanistic insight into this chloropentafluorosulfanylation system (Scheme 4). First, the introduction of a 2,2,6,6-tetramethyl-1-piperidinoxyl (TEMPO) effectively inhibited the reaction. Meanwhile, radical trapping studies using 1,1-diphenylethene further supported the formation of SF_5 and Cl radical (Scheme 4A). More compelling evidence for involvement of organic radicals was obtained from the radical clock experiment (Scheme 4B), where a ring-opened difunctionalization product **14** was detected in 27% yield, while only traces of the 1,2-addition product **13** were observed. This suggests that the steric hindrance at the 2-position in enynes plays a crucial role in the chlorine-atom transfer. In order to investigate the potential pathway of chlorine atom transfer, we investigated the effect of the 4-silyl group in 4-substituted 1,3-enynes on the regioselectivity. Unlike the aryl- or alkyl-substituted substrates, the chloropentafluorosulfanylation of compounds **15** proceeded in high total yields (78−83%) to give a mixture of 1,2- and 1,4-addition products. Notably, when the steric hindrance around the silicon atom is minimal (**15a** and **15b**), an ∼1:1 ratio of propargyl chlorides (**16a** and **16b**) and allenes (**16a′** and **16b′**) was observed. Conversely, with increased steric hindrance around the silicon atom (**15c**), the major product corresponds to the 1,2-addition (**16c**) with a ratio of 8:1. Since these silyl groups exhibit similar electron effects, we rationalized that this trend of regioselectivity likely arises from interactions between the fluorine atom in SF_5Cl and the silicon atom in the substrates during the intermolecular chlorine atom transfer from SF_5Cl to the carbon radical.

A plausible reaction pathway is proposed based on the above mechanistic experiment results and literature (Scheme 4D). The single electron transfer (SET) reaction between SF_5Cl and a Cu(I) complex gives a SF_5 radical and a Cu(II)−Cl species.[10] Subsequent addition of the SF_5 radical to 4-substituted 1,3-

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Scheme 4. Experimental Mechanistic Studies

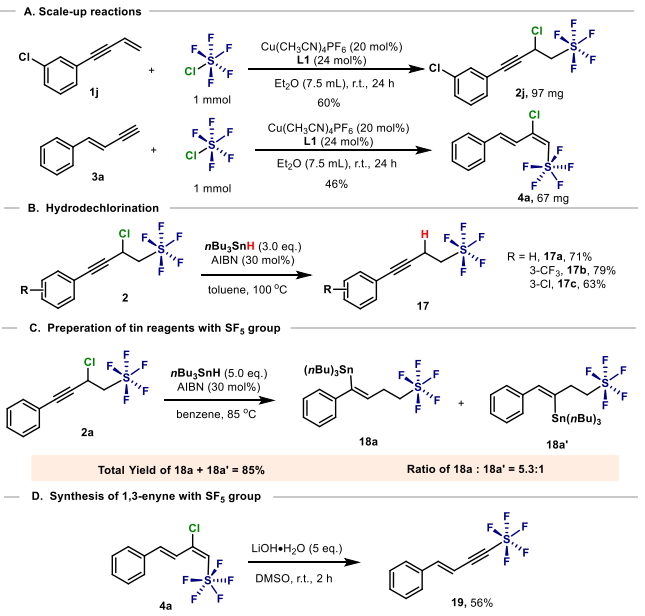


enyne gives a propargyl radical (**A**), which can isomerize to an allenyl radical (**A′**). Generally, Cl atom transfer occurred more rapidly than isomerization (**B**), resulting in the formation of 1,2-addition products. However, when a bulky substituent is present at the 2-position of enyne or when substituents at the 4-position interact with SF_5Cl, chlorine atom transfer preferentially occurs at the 4-position (**B′**), affording SF_5-substituted allenes.

Our direct pentafluorosulfanylation strategy demonstrated good synthetic values. Two reactions were scalable to a 1 mmol scale, giving the desired products in 60% and 46% yields for **2j** and **4a**, respectively (Scheme 5A). Furthermore, hydrodechlorination of **2**, in the presence of 2,2′-azobis(2-methylpropionitrile (AIBN) and tributyltin hydride,[17] resulted in high-yielding synthesis of the corresponding alkyne **17** (Scheme 5B). Increasing the loading of tributyltin hydride led to further hydrostannation of the alkyne[18] to give two regioisomeric olefins with a ratio of 5.3:1 (Scheme 5C). Alternatively, treatment of 1,3-diene **4a** with a strong base[19] provided the 4-SF_5-substituted 1,3-enyne via elimination of a HCl (Scheme 5D).

In conclusion, we have developed a copper-initiated regiodivergent chloropentafluorosulfanylation of 1,3-enynes

Scheme 5. Synthetic Applications



using SF_5Cl as the reagent. The regioselectivity of this system is dictated by the structural characteristics of the enyne substrates. Specifically, substrates containing a terminal olefin were selectively converted to the corresponding SF_5-substituted propargylic chlorides. For the 4-unsubstituted 1,3-enynes, only SF_5-substituted 1,3-dienes were obtained. When the substituents at the 2-position were bulkier than those at the 4-position, allenes bearing the SF_5 group became the major products. Regarding the reaction mechanism, the potential pathway of chlorine atom transfer from SF_5Cl to the carbon radical seems predominant. The novel chloropentafluorosulfanylation products can be transformed into various SF_5-containing building blocks through different methods, with potential applications in synthetic chemistry.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c01768.

Experimental procedures, characterization data, and spectra of new compounds (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) (a) Matsuzaki, K.; Okuyama, K.; Tokunaga, E.; Saito, N.; Shiro, M.; Shibata, N. Synthesis of diaryliodonium salts having pentafluorosulfanylarenes and their application to electrophilic pentafluorosulfanylarylation of C-, O-, N-, and S-nucleophiles. *Org. Lett.* **2015**, *17*, 3038−3041. (b) True, J. E.; Thomas, T. D.; Winter, R. W.; Gard, G. L. Electronegativities from core-ionization energies: electronegativities of SF_5 and CF_3. *Inorg. Chem.* **2003**, *42*, 4437−4441. (c) Sheppard, W. A. Arylsulfur pentafluorides. *J. Am. Chem. Soc.* **1962**, *84*, 3064−3072.

(2) Du, J.; Hua, G.; Beier, P.; Slawin, A. M.; Woollins, J. D. Single crystal X-ray structural features of aromatic compounds having a pentafluorosulfuranyl (SF_5) functional group. *Struct. Chem.* **2017**, *28*, 723−733.

(3) (a) Sani, M.; Zanda, M. Recent advances in the synthesis and medicinal chemistry of SF_5 and SF_4Cl compounds. *Synthesis* **2022**, *54*, 4184−4209. (b) Sowaileh, M. F.; Hazlitt, R. A.; Colby, D. A. Application of the pentafluorosulfanyl group as a bioisosteric replacement. *ChemMedChem.* **2017**, *12*, 1481−1490. (c) Wipf, P.; Mo, T.; Geib, S. J.; Caridha, D.; Dow, G. S.; Gerena, L.; Roncal, N.; Milner, E. E. Synthesis and biological evaluation of the first pentafluorosulfanyl analogs of mefloquine. *Org. Biomol. Chem.* **2009**, *7*, 4163−4165.

(4) (a) Jelínková, V.; Dellai, A.; Vachtlová, M.; Fecková, M.; Podlesný, J.; Klikar, M.; Castet, F.; Růžička, A.; Pařík, P.; Pytela, O. Molecular azo−imidazole photoswitches: Property tuning by substitution. *J. Photoch. Photobio. A* **2024**, *449*, 115390. (b) Sevilla, A. R.; Gao, H.; Steinberg, K. J.; Gallant, B. M. Elucidating Concentration-Dependent Energy Limitations in Li Primary Battery Fluoro-organosulfur Catholytes. *J. Phys. Chem. C* **2023**, *127*, 1722−1732.

(5) (a) Noonikara-Poyil, A.; Muñoz-Castro, A.; Boretskyi, A.; Mykhailiuk, P. K.; Dias, H. R. When SF5 outplays CF3: effects of pentafluorosulfanyl decorated scorpionates on copper. *Chem. Sci.* **2021**, *12*, 14618−14623. (b) Levin, M. D.; Ovian, J. M.; Read, J. A.; Sigman, M. S.; Jacobsen, E. N. Catalytic enantioselective synthesis of difluorinated alkyl bromides. *J. Am. Chem. Soc.* **2020**, *142*, 14831−14837.

(6) (a) Wang, L.; Ni, S.; Cornella, J. Validation of Arylphosphorothiolates as Convergent Substrates for Ar-SF_4Cl and Ar-SF_5 Synthesis. *Synthesis* **2021**, *53*, 4308−4312. (b) Pitts, C. R.; Bornemann, D.; Liebing, P.; Santschi, N.; Togni, A. Making the SF_5 Group More Accessible: A Gas-Reagent-Free Approach to Aryl Tetrafluoro-λ^6-sulfanyl Chlorides. *Angew. Chem., Int. Ed.* **2019**, *58*, 1950−1954. (c) Kanishchev, O. S.; Dolbier, Jr W. R. Synthesis and Characterization of 2-Pyridylsulfur Pentafluorides. *Angew. Chem., Int. Ed.* **2015**, *54*, 280−284.

(7) (a) Haufe, G. Synthesis and application of pentafluorosulfanylation reagents and derived aliphatic SF_5-containing building blocks. *Tetrahedron* **2022**, *109*, 132656. (b) Rombach, D.; Wagenknecht, H. A. Photoredox catalytic α-alkoxypentafluorosulfanylation of α-methyl- and α-phenylstyrene using SF_6. *Angew. Chem., Int. Ed.* **2020**, *59*, 300−303. (c) "Pentafluorosulfanyl chloride": Burton, D. J.; Wang, Y.; Bizet, V.; Cahard, D. *e-EROS Encyclopedia of Reagents for Organic Synthesis*; Wiley: Chichester, 2001.

(8) Gilbert, A.; Paquin, J.-F. Evaluation of the compatibility of pentafluorosulfanyl chloride with various solvents and additives. *J. Fluorine Chem.* **2019**, *221*, 70−74.

(9) (a) Kraemer, Y.; Bergman, E. N.; Togni, A.; Pitts, C. R. Oxidative fluorination of heteroatoms enabled by trichloroisocyanuric acid and potassium fluoride. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202205088. (b) Pitts, C. R.; Santschi, N.; Togni, A. *WO2019229103*, 2019.

(10) Shou, J. Y.; Xu, X. H.; Qing, F. L. Chemoselective hydro (chloro) pentafluorosulfanylation of diazo compounds with pentafluorosulfanyl chloride. *Angew. Chem., Int. Ed.* **2021**, *60*, 15271−15275.

(11) For selected exampels, see: (a) Nguyen, T. M.; Popek, L.; Matchavariani, D.; Blanchard, N.; Bizet, V.; Cahard, D. Expanding Radical Chloropentafluorosulfanylation of Alkynes. *Org. Lett.* **2024**, *26*, 365−369. (b) Ouellet-Du Berger, M.-R.; Boucher, M.; Paquin, J.-F. Light-activation of SF_5Cl for the atom transfer radical addition onto alkenes and alkynes. *J. Fluorine Chem.* **2023**, *268*, 110131. (c) Birepinte, M.; Champagne, P. A.; Paquin, J. F. Photoinitiated anti-Hydropentafluorosulfanylation of Terminal Alkynes. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202112575. (d) Gilbert, A.; Birepinte, M.; Paquin, J.-F. Electron donor-acceptor (EDA)-complex enabled SF_5Cl addition on alkenes and alkynes. *J. Fluorine Chem.* **2021**, *243*, 109734. (e) Ait-Mohand, S.; Dolbier, W. R. New and convenient method for incorporation of pentafluorosulfanyl (SF_5) substituents into aliphatic organic compounds. *Org. Lett.* **2002**, *4*, 3013−3015.

(12) Shou, J. Y.; Qing, F. L. Three-Component Reaction of Pentafluorosulfanyl Chloride, Alkenes and Diazo Compounds and Synthesis of Pentafluorosulfanylfurans. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202208860.

(13) (a) Rombach, D.; Birenheide, B.; Wagenknecht, H. A. Photoredox Catalytic Pentafluorosulfanylative Domino Cyclization of α-Substituted Alkenes to Oxaheterocycles by Using SF_6. *Chem.—Eur. J.* **2021**, *27*, 8088−8093. (b) Feng, F.-F.; Ma, J.-A.; Cahard, D. Radical 1, 5-chloropentafluorosulfanylation of unactivated vinylcyclopropanes and transformation into α-SF_5 ketones. *J. Org. Chem.* **2021**, *86*, 13808−13816.

(14) For selected exampels, see: (a) Zhang, Z.; Gevorgyan, V. Escape from Hydrofunctionalization: Palladium Hydride-Enabled Difunctionalization of Conjugated Dienes and Enynes. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202311848. (b) Wang, B.-R.; Li, Y.-B.; Zhang, Q.; Gao, D.; Tian, P.; Li, Q.; Yin, L. Copper (I)-catalyzed asymmetric 1, 3-dipolar cycloaddition of 1, 3-enynes and azomethine ylides. *Nat. Commun.* **2023**, *14*, 4688. (c) Song, H.; Zhang, X.; Chen, G.; He, X.; Lian, Z. Copper-Catalyzed 1, 4-Trifluoromethylthio-Arylsulfonylation of 1, 3-Enynes via the Insertion of Sulfur Dioxide. *Org. Lett.* **2023**, *25*, 5916−5921. (d) Dong, X. Y.; Zhan, T. Y.; Jiang, S. P.; Liu, X. D.; Ye, L.; Li, Z. L.; Gu, Q. S.; Liu, X. Y. Copper-catalyzed asymmetric coupling of allenyl radicals with terminal alkynes to access tetrasubstituted allenes. *Angew. Chem., Int. Ed.* **2021**, *60*, 2160−2164.

(15) (a) Li, Y.; Bao, H. Radical transformations for allene synthesis. *Chem. Sci.* **2022**, *13*, 8491−8506. (b) Zhang, K. F.; Bian, K. J.; Li, C.; Sheng, J.; Li, Y.; Wang, X. S. Nickel-catalyzed carbofluoroalkylation of 1, 3-enynes to access structurally diverse fluoroalkylated allenes. *Angew. Chem., Int. Ed.* **2019**, *58*, 5069−5074. (c) Zhu, X.; Deng, W.; Chiou, M.-F.; Ye, C.; Jian, W.; Zeng, Y.; Jiao, Y.; Ge, L.; Li, Y.; Zhang, X.; Bao, H. Copper-catalyzed radical 1, 4-difunctionalization of 1, 3-enynes with alkyl diacyl peroxides and N-fluorobenzenesulfonimide. *J. Am. Chem. Soc.* **2019**, *141*, 548−559.

(16) Wang, F.; Wang, D.; Zhou, Y.; Liang, L.; Lu, R.; Chen, P.; Lin, Z.; Liu, G. Divergent synthesis of CF_3-substituted allenyl nitriles by

ligand-controlled radical 1, 2-and 1, 4-addition to 1, 3-enynes. *Angew. Chem., Int. Ed.* **2018**, *57*, 7140−7145.
(17) Bachi, M. D.; De Mesmaeker, A.; Stevenart-De Mesmaeker, N. Free-radical annelation in the synthesis of bicyclic *β*-lactams. 5. Synthesis of carbapenams. *Tetrahedron Lett.* **1987**, *28*, 2637−2640.
(18) Alami, M.; Hamze, A.; Provot, O. Hydrostannation of alkynes. *ACS Catal.* **2019**, *9*, 3437−3466.
(19) Popek, L.; Cihan, M.; Blanchard, N.; Bizet, V. Palladium-Catalyzed Regioselective Synthesis of 2-SF5-Indenols and Further Derivatizations. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e202315909.

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